Principles of Geographic Information Systems

MODULE-1: A gentle introduction to GIS, Geographic Information and Spatial Database

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Certificate

This is to certify that the e-book titled "PRINCIPLES OF GEOGRAPHIC INFORMATION SYSTEMS" comprises all elementary learning tools for a better understating of the relevant concepts. This e-book is comprehensively compiled as per the predefined eight parameters and guidelines.

PushpaM

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Sem-6 Principles of Geographic Information Systems

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Recommended Books:

1. Principles of Geographic Information Systems: An introductory textbook, Editors: Otto Huisman and Rolf A.
2. Principles of Geographical Information Systems – Peter A. Burrough (Author), Rachael A. McDonnell (Author), Christopher D. Lloyd (Author)

- **Prerequisites and Linking:**

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Chapter 01

A gentle introduction to GIS

1.1 The nature of GIS

The acronym GIS stands for geographic information system. As the name suggests, a GIS is a tool for working with geographic information. GIS have rapidly developed since the late 1970's in terms of both technical and processing capabilities, and today are widely used all over the world for a wide range of purposes. Let us begin by looking at some of these:

• An urban planner might want to assess the extent of urban fringe growth in her/his city, and quantify the population growth that some suburbs are witnessing. S/he might also like to understand why these particular suburbs are growing and others are not.

• A biologist might be interested in the impact of slash-and-burn practices on the populations of amphibian species in the forests of a mountain range to obtain a better understanding of long-term threats to those populations.

• A natural hazard analyst might like to identify the high-risk areas of annual monsoon-related flooding by investigating rainfall patterns and terrain characteristics.

• A geological engineer might want to identify the best localities for constructing buildings in an earthquake-prone area by looking at rock formation characteristics;

• A mining engineer could be interested in determining which prospective copper mines should be selected for future exploration, taking into account parameters such as extent, depth and quality of the ore body, amongst others;

• A geoinformatics engineer hired by a telecommunications company may want to determine the best sites for the company's relay stations, taking into account various cost factors such as land prices, undulation of the terrain

• A forest manager might want to optimize timber production using data on soil and current tree stand distributions, in the presence of a number of operational constraints, such as the need to preserve species diversity in the area;

• A hydrological engineer might want to study a number of water quality parameters of different sites in a freshwater lake to improve understanding of the current distribution of Typha reed beds, and why it differs from that of a decade ago.

Additional Learning: Additional uses of GIS:
https://mapasyst.extension.org/what-are-some-uses-for-gis/#:~:text=Common%20uses%20of%20GIS%20include,and%20managing%20agricultural%20crop%20data%2C

1.1.1 Some fundamental observations

Our world is dynamic. Many aspects of our daily lives and our environment are constantly changing, and not always for the better. Some of these changes appear to have natural causes (e.g. volcanic eruptions, meteorite impacts), while others are the result of human modification of the environment (e.g. land use changes or land reclamation from the sea, a favourite pastime of the Dutch).

In order to understand what is going on in our world, we study the processes or phenomena that bring about geographic change. In many cases, we want to broaden or deepen our understanding to help us make decisions, so that we can take the best course of action. For instance, if we understand El Ni˜ no better, and can forecast that another event may take place in the year 2012, we can devise an action plan to reduce the expected losses in the fishing industry, to lower the risks of landslides caused by heavy rains or to build up water supplies in areas of expected droughts.

Video on What is GIS, https://www.youtube.com/watch?v=-ZFmAAHBfOU

1.1.2 Defining GIS:

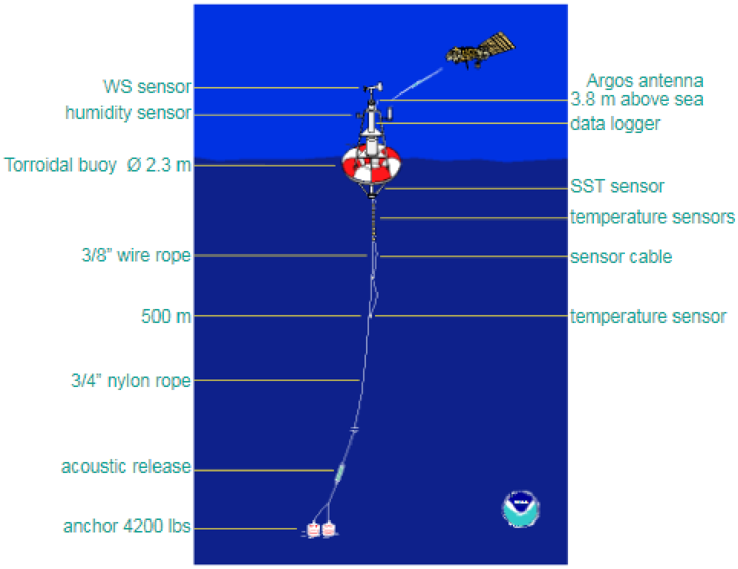
A GIS is a computer-based system that provides the following four sets of capabilities to handle georeferenced data:

1. Data capture and preparation
2. Data management, including storage and maintenance
3. Data manipulation and analysis
4. Data presentation

This implies that a GIS user can expect support from the system to enter (georeferenced) data, to analyse it in various ways, and to produce presentations (including maps and other types) from the data. This would include support for various kinds of coordinate systems and transformations between them, options for analysis of the georeferenced data, and obviously a large degree of freedom of choice in the way this information is presented (such as colour scheme, symbol set, and medium used).

Data capture and preparation: In the El Ni˜no case, data capture refers to the collection of sea water temperatures and wind speed measurements. This is achieved by placing buoys with measuring equipment at various places in the ocean. Each buoy measures a number of things: wind speed and direction; air temperature and humidity; and sea water temperature at the surface and at various depths down to 500 metres. For the sake of our example we will focus on sea surface temperature (SST) and wind speed (WS).

A typical buoy is illustrated in Figure, which shows the placement of various sensors on the buoy. For monitoring purposes, some 70 buoys were deployed at strategic places within 10◦ latitude of the Equator, between the Galapagos Islands and Papua New Guinea.



Data management:

For our example application, data management refers to the storage and maintenance of the data transmitted by the buoys via satellite communication. This phase requires a decision to be made on how best to represent our data, both in terms of their spatial properties and the various attribute values which we need to store.

Data manipulation and analysis:

It appears that the following steps took place for the upper two figures:

1. For each buoy, the average SST for each month was computed, using the daily SST measurements for that month. This is a simple computation.
2. For each buoy, the monthly average SST was taken together with the geographic location, to obtain a georeferenced list of averages.

3. From this georeferenced list, through a method of spatial interpolation, the estimated SST of other positions in the study are were computed. This step was performed as often as needed, to obtain a fine mesh of positions with measured or estimated SSTs from which the maps of Figure were eventually derived.
4. We assume that previous to the above steps we had obtained data about average SST for the month of December for a series of years. This too may have been spatially interpolated to obtain a 'normal situation' December data set of a fine resolution.

Data presentation:

After the data manipulations discussed above, our data is prepared for producing output. The data presentation phase deals with putting it all together into a format that communicates the result of data analysis in the best possible way.

Steps:

• The message we wanted to portray is what are the El Ni˜no and La Ni˜na events, both in absolute figures, but also in relative figures, i.e. as differences from a normal situation.
• The audience for this data presentation clearly were the readers of this text book.
• The medium was this book, (printed matter of A4 size) and possibly a website.
• The rules of aesthetics demanded many things: the maps should be printed north-up; with clear georeferencing; with intuitive use of symbols et cetera.
• The techniques that we used included the use of a colour scheme and isolines, plus a number of other techniques.

Video on El nino: https://www.youtube.com/watch?v=d6s0T0m3F8s

1.1.3 GISystems, GIScience and GIS applications:

Functioning GIS requires both hardware and software, and also people such as GISystems the database creators or administrators, analysts who work with the software, and the users of the end product.

Geo-Information Science is the scientific field that attempts to integrate different disciplines studying the methods and techniques of handling spatial information. Related terms include geoinformatics, geomatics, and spatial information science. These are all similar terms which have much the same meaning, although each GIScience approach has slight differences in the way it deals with problems, some emphasizing engineering approaches, others computational solutions, and so on.

GIS software can (generically) be applied to many different applications. When there is no risk of ambiguity, people sometimes do not make the distinction between a 'GIS' and a 'GIS application'. Project-based GIS applications usually have a clear-cut purpose, and these applications can be short-lived: the research is carried out by collecting data, entering data in the GIS, analysing the data, and producing informative maps. An example is rapid earthquake damage assessment. Institutional GIS applications, on the other hand, usually have as their goal the continued administration of spatial change and the sustained availability of spatial base data.

Additional Learning: Research paper on Future development of GIScience and GISystems:

https://www.researchgate.net/publication/316780924_The_Future_Development_of_GISystems_GIScience_and_GIServices

1.1.4 Spatial data and geoinformation

By data, we mean representations that can be operated upon by a computer. More specifically, by spatial data we mean data that contains positional values, such as (x,y) co-ordinates. Sometimes the more precise phrase geospatial data is used as a further refinement, which refers to spatial data that is georeferenced. By Geospatial data and geoinformation information, we mean data that has been interpreted by a human being. Humans work with and act upon information, not data. Human perception and mental processing leads to information, and hopefully understanding and knowledge. Geoinformation is a specific type of information resulting from the interpretation of spatial data.

National Mapping Agencies (NMAs), which are responsible for collecting topographic data for the entire country following pre-set standards.

Other agencies such as geological survey companies, energy supply companies, local government departments, and many others, all collect and maintain spatial data for their own particular purposes. If data is to be shared among different users, these users need to know not only what data exists, where and in what format it is held, but also whether the data meets their particular quality requirements. This 'data about data' is known as metadata.

Key components of spatial data quality include positional accuracy (both horizontal and vertical), temporal accuracy (that the data is up to date), attribute accuracy (e.g. in labelling of features or of classifications), lineage (history of the data including sources), completeness (if the data set represents all related features of reality), and logical consistency (that the data is logically structured).

These components play an important role in assessment of data quality for several reasons:

1. Even when source data, such as official topographic maps, have been subject to stringent quality control, errors are introduced when these data are input to GIS.
2. Unlike a conventional map, which is essentially a single product, a GIS database normally contains data from different sources of varying quality.
3. Unlike topographic or cadastral databases, natural resource databases contain data that are inherently uncertain and therefore not suited to conventional quality control procedures.
4. Most GIS analysis operations will themselves introduce errors.

Video on Spatial Data: https://www.youtube.com/watch?v=O7kg3dQ8WDY

1.2 The real world and representations of it:

One of the main uses of GIS is as a tool to help us make decisions. Specifically, we often want to know the best location for a new facility, the most likely sites for mosquito habitat, or perhaps identify areas with a high risk of flooding so that we can formulate the best policy for prevention. In using GIS to help make these decisions, we need to represent some part of the real world as it is, as it was, or perhaps as we think it will be. We need to restrict ourselves to 'some part' of the real world simply because it cannot be represented completely.

1.2.1 Models and modelling

'Modelling' is a term used in many different ways and which has many different meanings. A representation of some part of the real world can be considered a model because the representation will have certain characteristics in common with the real world. This then allows us to study and operate on the model itself instead of the real world in order to test what happens under various conditions, and help us answer 'what if' questions. We can change the data or alter the parameters of the model, and investigate the effects of the changes.

Models—as representations—come in many different flavours. In the GIS environment, the most familiar model is that of a map. A map is a miniature representation of some part of the real world. A database can store a considerable amount of data, and also provides various functions to operate on the stored data. The collection of stored data represents some real world phenomena, so it too is a model. Obviously, here we are especially interested in databases that store spatial data. Digital models (as in a database or GIS) have enormous advantages over paper models (such as maps). They are more flexible, and therefore more easily changed for the purpose at hand. In principle, they allow animations and simulations to be carried out by the computer system. This has opened up an important toolbox that can help to improve our understanding of the world.

Video on Models and modelling: https://www.youtube.com/watch?v=-ZFmAAHBfOU

1.2.2 Maps: Maps have been used for thousands of years to represent information about the real world, and continue to be extremely useful for many applications in various domains. Their conception and design has developed into a science with a high degree of sophistication. A disadvantage of the traditional paper map is that it is generally restricted to two-dimensional static representations, and that it is always displayed in a fixed scale. Cartography, as the science and art of map making, functions as an interpreter, translating real world phenomena (primary data) into correct, clear and understandable representations for our use. Maps also become a data source for other applications, including the development of other maps.
With the advent of computer systems, analogue cartography developed into digital cartography, and computers play an integral part in modern cartography.
Alongside this trend, the role of the map has also changed accordingly, and the dominance of paper maps is eroding in today's increasingly 'digital' world.

1.2.3 Databases

A database is a repository for storing large amounts of data. It comes with a number of useful functions:

1. A database can be used by multiple users at the same time—i.e. it allows concurrent use,
2. A database offers a number of techniques for storing data and allows the use of the most efficient one—i.e. it supports storage optimization,
3. A database allows the imposition of rules on the stored data; rules that will be automatically checked after each update to the data—i.e. it supports data integrity,
4. A database offers an easy to use data manipulation language, which allows the execution of all sorts of data extraction and data updates—i.e. it has a query facility,
5. A database will try to execute each query in the data manipulation language in the most efficient way—i.e. it offers query optimization.

1.2.4 Spatial databases and spatial analysis

A GIS must store its data in some way. For this purpose the previous generation of software was equipped with relatively rudimentary facilities. Since the 1990's there has been an increasing trend in GIS applications that used a GIS for spatial analysis, and used a database for storage. In more recent years, spatial databases (also known as geodatabases) have emerged. Besides traditional administrative data, they can store representations of real world geographic phenomena for use in a GIS. These databases are special because they use additional techniques different from tables to store these spatial representations.
A geodatabase is not the same thing as a GIS, though both systems share a number of characteristics. These include the functions listed above for databases in general:

concurrency, storage, integrity, and querying, specifically, but not only, spatial data. A GIS, on the other hand, is tailored to operate on spatial data.

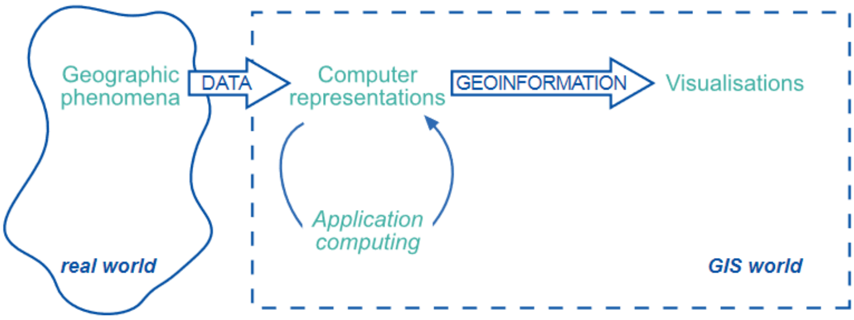
This is probably GIS's main strength: providing various ways to combine representations of geographic phenomena. GISs has built-in tools for map production, of the paper and the digital kind. They operate with an 'embedded understanding' of geographic space. Databases typically lack this kind of understanding.

Spatial analysis is the generic term for all manipulations of spatial data carried out to improve one's understanding of the geographic phenomena that the data represents. It involves questions about how the data in various layers might relate to each other, and how it varies over space.

Chapter 2
Geographic information and Spatial data types

2.1 Models and representations of the real world:

Modelling is the process of producing an abstraction of the 'real world' so that some part of it can be more easily handled.
Depending on the application domain of the model, it may be necessary to manipulate the data with specific techniques. To investigate the geology of an area, we may be interested in obtaining a geological classification. This may result in additional computer representations, again stored in bits and bytes. To examine how the data is stored inside the GIS, one could look into the actual data files, but this information is largely meaningless to a normal user.



In order to better understand both our representation of the phenomena, and our eventual output from any analysis, we can use the GIS to create visualizations from the computer representation, either on-screen, printed on paper, or otherwise. It is crucial to understand the fundamental differences between these notions. The real world, after all, is a completely different domain than the 'GIS' world, in which we build models or simulations of the real world.
Given the complexity of real world phenomena, our models can by definition never be perfect. We have limitations on the amount of data that we can store, limits on the amount of detail we can capture, and (usually) limits on the time we have available for a project.
Any geographic phenomenon can usually be represented in various ways; the choice of which representation is best depends mostly on two issues. Firstly, what original, raw data (from sensors or otherwise) is available, and secondly, what sort of data manipulation is required or will be undertaken.

2.2 Geographic phenomena:

2.2.1 Defining geographic phenomena

A GIS operates under the assumption that the relevant spatial phenomena occur in a two- or three-dimensional Euclidean space, unless otherwise specified. Euclidean space can be informally defined as a model of space in which locations

Euclidean space are represented by coordinates— (x,y) in 2D; (x,y,z) in 3D—and Distance and direction can defined with geometric formulas. In the 2D case, this is known as the Euclidean plane, which is the most common Euclidean space in GIS use. In order to be able to represent relevant aspects real world phenomena inside a GIS, we first need to define what it is we are referring to. We might define a geographic phenomenon as a manifestation of an entity or process of interest that:

• Can be named or described,
• Can be georeferenced, and
• Can be assigned a time (interval) at which it is/was present.

The relevant phenomena for a given application depends entirely on one's objectives. For instance, in water management, the objects of study might be river basins, agro-ecologic units, measurements of actual evapotranspiration, meteorological data, ground water levels, irrigation levels, water budgets and measurements of total water use.

Additional Learning: Case study on Geographic Phenomena https://gistbok.ucgis.org/bok-topics/perception-and-cognitive-processing-geographic-phenomena

2.2.2 Types of geographic phenomena

The attempted definition of geographic phenomena above is necessarily abstract, and therefore perhaps somewhat difficult to grasp. The main reason for this is that geographic phenomena come in so many different 'flavours', which we will try to categorize below. Before doing so, we must make two further observations.

Firstly, In order to be able to represent a phenomenon in a GIS requires us to state what it is, and where it is. We must provide a description—or at least a name—on the one hand, and a georeference on the other hand.

Secondly, some phenomena manifest themselves essentially everywhere in the study area, while others only do so in certain localities. If we define our study area as the equatorial Pacific Ocean, we can say that Sea Surface Temperature can be measured anywhere in the study area. Therefore, it is a typical example of a (geographic) field.

(Geographic) objects populate the study area, and are usually well-distinguished, discrete, and bounded entities. The space between them is potentially 'empty' or undetermined.

2.2.3 Geographic fields

A field is a geographic phenomenon that has a value 'everywhere' in the study area. We can therefore think of a field as a mathematical function f that associates a specific value with any position in the study area. Hence if (x,y) is a position in the study area, then f(x,y) stands for the value of the field f at locality (x,y).

Fields can be discrete or continuous. In a continuous field, the underlying function is assumed to be 'mathematically smooth', meaning that the field values along any path through the study area do not change abruptly, but only gradually.
A continuous field can even be differentiable, meaning we can determine a measure of change in the field value per unit of distance anywhere and in any direction. For example, if the field is elevation, this measure would be slope, i.e. the change of elevation per metre distance; if the field is soil salinity, it would be salinity gradient, i.e. the change of salinity per metre distance. Discrete fields divide the study space in mutually exclusive, bounded parts, with all locations in one part having the same field value. Typical examples are land classifications, for instance, using either geological classes, soil type, land use type, crop type or natural vegetation type.

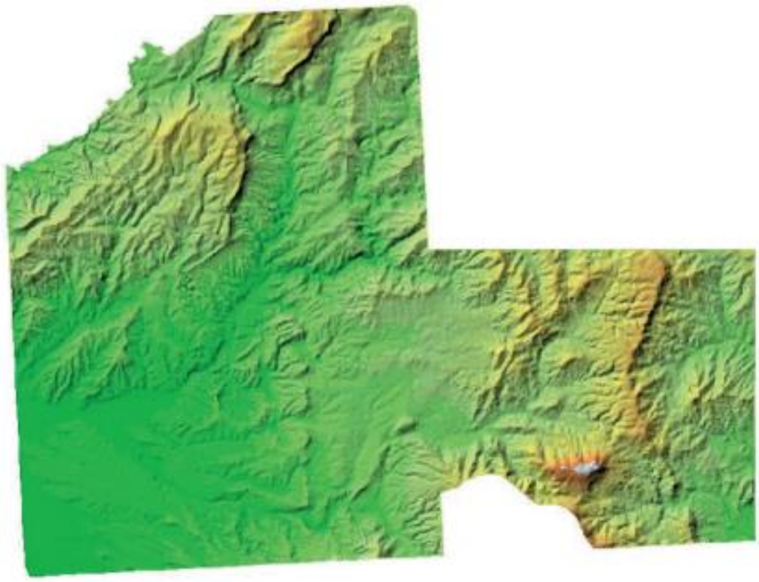
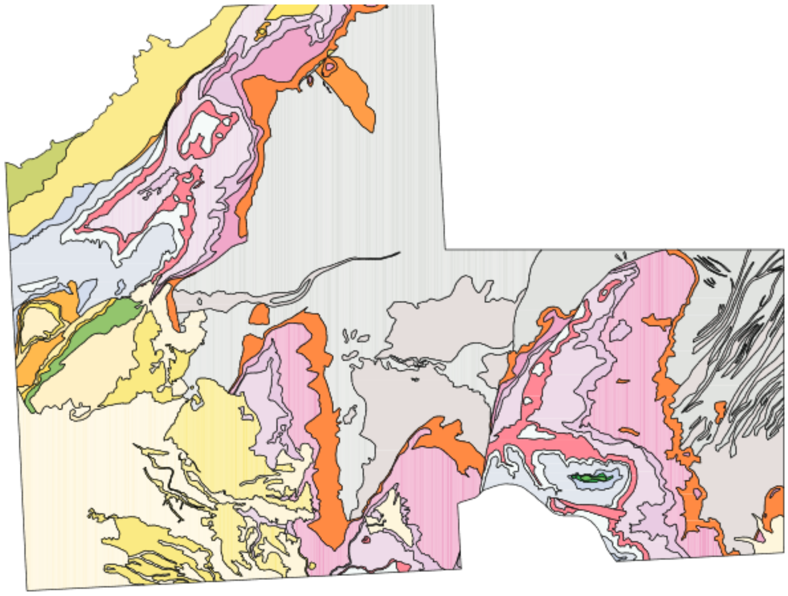


Figure 2.2: A continuous field example, namely the *elevation* in the study area of Falset, Spain.
Data source: Department of Earth Systems Analysis (ESA, ITC)

Essentially, these two types of fields differ in the type of cell values. A discrete field like land use type will store cell values of the type 'integer'. Therefore it is also called an integer raster. Discrete fields can be easily converted to polygons, since it is relatively easy to draw a boundary line around a group of cells with the same value. A continuous raster is also called a 'floating point' raster. A field-based model consists of a finite collection of geographic fields: we may be interested in elevation, barometric pressure, mean annual rainfall, and maximum daily evapotranspiration, and thus use four different fields to model the relevant phenomena within our study area.



Data types and values:

1. Nominal data values are values that provide a name or identifier so that we can discriminate between different values, but that is about all we can do. Specifically, we cannot do true computations with these values. An example is the names of geological units. This kind of data value is called categorical data when the values assigned are sorted according to some set of non-overlapping categories. For example, we might identify the soil type of a given area to belong to a certain (pre-defined) category.
2. Ordinal data values are data values that can be put in some natural sequence but that do not allow any other type of computation. Household income, for instance, could be classified as being either 'low', 'average' or 'high'. Clearly this is their natural sequence, but this is all we can say—we can not say that a high income is twice as high as an average income.
3. Interval data values are quantitative, in that they allow simple forms of computation like addition and subtraction. However, interval data has no arithmetic zero value, and does not support multiplication or division. For instance, a temperature of 20◦ C is not twice as warm as 10◦C, and thus centigrade temperatures are interval data values, not ratio data values.
4. Ratio data values allow most, if not all, forms of arithmetic computation.

Additional Learning: Datatypes: https://support.esri.com/en/other-resources/gis-dictionary/term/dcda78ea-3344-4276-a31b-9ace36c0fb43

2.2.4 Geographic objects:

When a geographic phenomenon is not present everywhere in the study area, but somehow 'sparsely' populates it, we look at it as a collection of geographic objects.

Such objects are usually easily distinguished and named, and their position in space is determined by a combination of one or more of the following parameters:

- Location (where is it?),
- Shape (what form is it?),
- Size (how big is it?), and
- Orientation (in which direction is it facing?).

How we want to use the information about a geographic object determines which of the four above parameters is required to represent it. For instance, in an in-car navigation system, all that matters about geographic objects like petrol stations is where they are. Thus, location alone is enough to describe them in this particular context, and shape, size and orientation are not necessarily relevant.

In the same system, however, roads are important objects, and for these some notion of location (where does it begin and end), shape (how many lanes does it have), size (how far can one travel on it) and orientation (in which direction can one travel on it) seem to be relevant information components.

Shape is usually important because one of its factors is dimension. This relates to whether an object is perceived as a point feature, or a linear, area or volume feature. The petrol stations mentioned above apparently are zero-dimensional, i.e. they are perceived as points in space; roads are one-dimensional, as they are considered to be lines in space. In another use of road information—for instance, in multi-purpose cadastre systems where precise location of sewers and manhole covers matters—roads might well be considered to be two-dimensional entities, i.e. areas within which a manhole cover may fall.

Collections of geographic objects can be interesting phenomena at a higher aggregation level: forest plots form forests, groups of parcels form suburbs, streams, brooks and rivers form a river drainage system, roads form a road network, and SST buoys form an SST sensor network. It is sometimes useful to view geographic phenomena at this more aggregated level and look at characteristics like coverage, connectedness, and capacity. For example:

- Which part of the road network is within 5 km of a petrol station? (A coverage question)
- What is the shortest route between two cities via the road network? (A connectedness question)
- How many cars can optimally travel from one city to another in an hour? (A capacity question)

2.2.5 Boundaries:

Location, shape and size are fully determined if we know an area's boundary, so the boundary is a good candidate for representing it. This is especially true for areas that have naturally crisp boundaries. A crisp boundary is one that can be determined with almost arbitrary precision, dependent only on the data acquisition technique applied. Fuzzy boundaries contrast with crisp boundaries in that the boundary is not a precise line, but rather itself an area of transition.

As a general rule-of-thumb, crisp boundaries are more common in man-made phenomena, whereas fuzzy boundaries are more common with natural phenomena. In recent years, various research efforts have addressed the issue of explicit treatment of fuzzy boundaries, but there is still limited support for these in existing GIS software.

2.3 Computer representations of geographic information

Elevation, for instance, can be measured at many locations, even within one's own backyard, and each location may give a different value. In order to represent such a phenomenon faithfully in computer memory, we could either:

• Try to store as many (location, elevation) observation pairs as possible, or

• Try to find a symbolic representation of the elevation field function, as a formula in x and y —like ($3.0678x^2 + 20.08x - 7.34y$) or so—which can be evaluated to give us the elevation at any given (x,y) location.

Both of these approaches have their drawbacks. The first suffers from the fact that we will never be able to store all elevation values for all locations; after all, there are infinitely many locations. The second approach suffers from the fact that we do not know just what this function should look like, and that it would be extremely difficult to derive such a function for larger areas. In GISs, typically a combination of both approaches is taken. We store a finite, but intelligently chosen set of (sample) locations with their elevation. This gives us the elevation for those stored locations, but not for others.

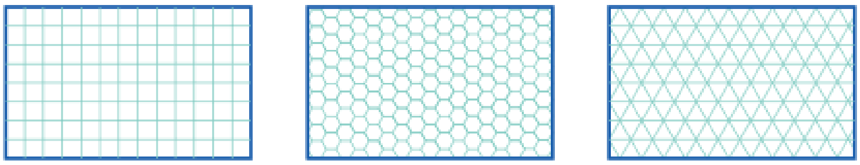
Interpolation is made possible by a principle called spatial autocorrelation. This is a fundamental principle which refers to the fact that locations that are closer together are more likely to have similar values than locations that are far apart — commonly referred to as 'Tobler's first law of Geography'. An obvious example of a phenomenon which exhibits this property is sea-surface temperature, where one might expect a high degree of correlation between measures taken close together.

Line objects, either by themselves or in their role of region object boundaries, are another common example of continuous phenomena that must be finitely represented. In real life, these objects are usually not straight, and are often erratically curved.

From this it becomes clear that phenomena with intrinsic continuous and/or infinite characteristics have to be represented with finite means (computer memory) for computer manipulation, and any finite representation scheme is open to errors of interpretation. To this end, fields are usually implemented with a tessellation approach, and objects with a (topological) vector approach. However, this is not a hard-and-fast rule, as practice sometimes demands otherwise.

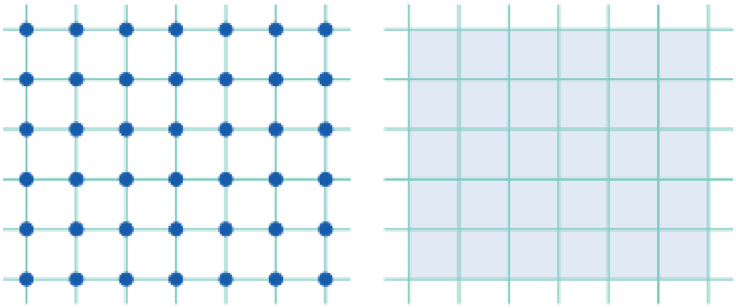
2.3.1 Regular tessellations

A tessellation (or tiling) is a partitioning of space into mutually exclusive cells that together make up the complete study space. With each cell, some (thematic) value is associated to characterize that part of space. Three regular tessellation types are illustrated in Figure. In a regular tessellation, the cells are the same shape and size. The simplest example is a rectangular raster of unit squares, represented in a computer in the 2D case as an array of $n \times m$ elements.



In all regular tessellations, the cells are of the same shape and size, and the field attribute value assigned to a cell is associated with the entire area occupied by the cell. The square cell tessellation is by far the most commonly used, mainly because georeferencing a cell is so straightforward. These tessellations are known under various names in different GIS packages, but most frequently as rasters.

A raster is a set of regularly spaced (and contiguous) cells with associated (field) values. The associated values represent cell values, not point values. This means that the value for a cell is assumed to be valid for all locations within the cell.
The size of the area that a single raster cell represents is called the raster’s resolution. Sometimes, the word grid is also used, but strictly speaking, a grid refers to values at the intersections of a network of regularly spaced horizontal and perpendicular lines



The field value of a cell can be interpreted as one for the complete tessellation cell, in which case the field is discrete, not continuous or even differentiable.
To improve on this continuity issue, we can do two things:

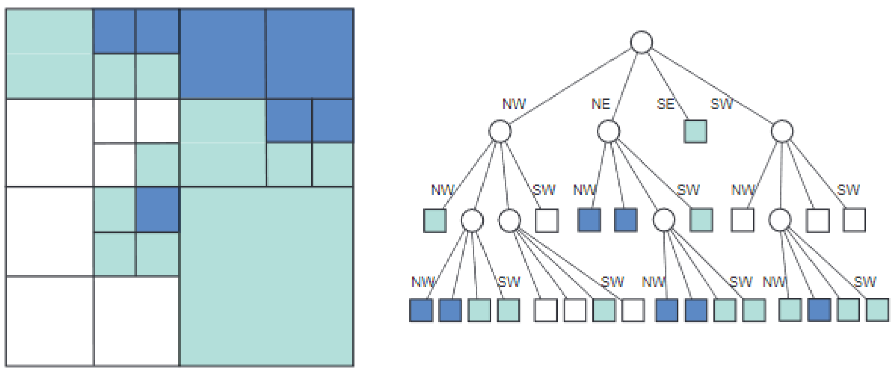
•Make the cell size smaller, so as to make the 'continuity gaps' between the cells smaller, and/or

•Assume that a cell value only represents elevation for one specific location in the cell, and to provide a good interpolation function for all other locations that has the continuity characteristic.

The location associated with a raster cell is fixed by convention, and may be the cell centroid (mid-point) or, for instance, its left lower corner. Values for other positions than these must be computed through some form of interpolation function, which will use one or more nearby field values to compute the value at the requested position. This allows us to represent continuous, even differentiable, functions.
An important advantage of regular tessellations is that we know how they partition space, and we can make our computations specific to this partitioning. This leads to fast algorithms.

2.3.2 Irregular tessellations

These are partitions of space into mutually disjoint cells, but now the cells may vary in size and shape, allowing them to adapt to the spatial phenomena that they represent. A well-known data structure in this family—upon which many more variations have been based—is the region quad tree. It is based on a regular tessellation of square cells, but takes advantage of cases where neighbouring cells have the same field value, so that they can together be represented as one bigger cell. A simple illustration is provided in Figure. It shows a 8×8 raster with three possible field values: white, green and blue. The quad tree that represents this raster is constructed by repeatedly splitting up the area into four quadrants, which are called NW, NE, SE, SW for obvious reasons. This procedure stops when all the cells in a quadrant have the same field value. The procedure produces an upside-down, tree-like structure, known as a quad tree.



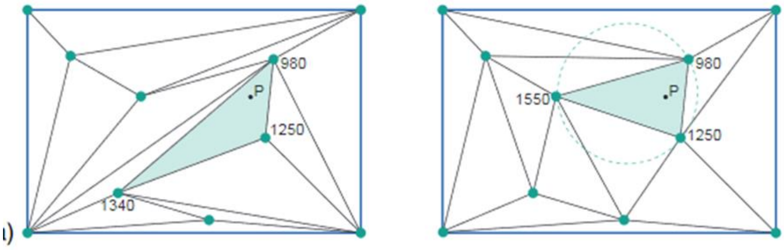
Quadtrees are adaptive because they apply the spatial auto correlation principle, i.e. that locations that are near in space are likely to have similar field values. When a conglomerate of cells has the same value, they are represented together in the quadtree, provided boundaries coincide with the predefined quadrant boundaries.

Video on Quadtree: https://www.youtube.com/watch?v=fA1Omsl1msU

2.3.3 Vector representations

In vector representations, an attempt made to explicitly associate georeferences with the geographic phenomena. A georeferences explicitly is a coordinate pair from some geographic space, and is also known as a vector. A representation for geographic fields can be considered a hybrid between tessellations and vector representation.

Triangulated Irregular Networks: A commonly used data structure in GIS software is the triangulated irregular network, or TIN. It is one of the standard implementation techniques for digital terrain models, but it can be used to represent any continuous field. The principles behind a TIN are simple. It is built from a set of locations for which we have a measurement, for instance an elevation. The locations can be arbitrarily scattered in space, and are usually not on a nice regular grid. Any location together with its elevation value can be viewed as a point in three-dimensional space. From these 3D points, we can construct an irregular tessellation made of triangles. Two such tessellations are illustrated in Figure.



Some tessellations are better than others, in the sense that they make smaller errors of elevation approximation. For instance, if we base our elevation computation for location P on the left hand shaded triangle, we will get another value than from the right hand shaded triangle.

A TIN clearly is a vector representation: each anchor point has a stored georeference. Yet, we might also call it an irregular tessellation, as the chosen triangulation provides a partitioning of the entire study space. However, in this case, the cells do not have an associated stored value as is typical of tessellations, but rather a simple interpolation function that uses the elevation values of its three anchor points.

Video in TIN: https://www.youtube.com/watch?v=15alCLUKrL4&t=84s

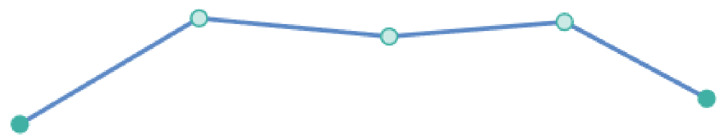
Point representations:

Points are used to represent objects that are best described as shape- and size-less, one-dimensional features. Whether this is the case really depends on the purposes of the spatial application and also on the spatial extent of the objects compared to the scale applied in the application. For a tourist city map, a park will not usually be considered a point feature, but perhaps a museum will, and certainly a public phone booth might be represented as a point.

Besides the georeference, usually extra data is stored for each point object. This so-called attribute or thematic data, can capture anything that is considered relevant about the object. For phone booth objects, this may include the owning telephone company, the phone number, or the data last serviced.

Line representations:

Line data are used to represent one-dimensional objects such as roads, rail roads, canals, rivers and power lines. Again, there is an issue of relevance for the application and the scale that the application requires. For the example application of mapping tourist information, bus, subway and streetcar routes are likely to be relevant line features. Some cadastral systems, on the other hand, may consider roads to be two-dimensional features, i.e. having a width as well.

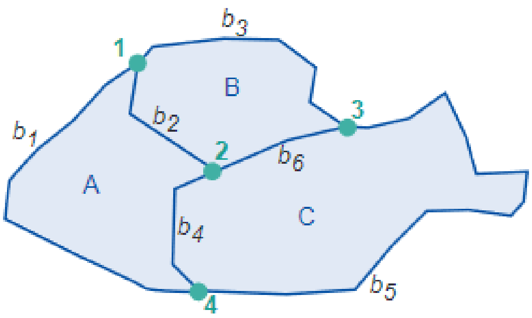


Collections of (connected) lines may represent phenomena that are best viewed as networks. With networks, specific types of interesting questions arise that have to do with connectivity and network capacity. These relate to applications such Networks as traffic monitoring and watershed management. With network elements—i.e. the lines that make up the network—extra values are commonly associated like distance, quality of the link, or carrying capacity.

Area representations:

When area objects are stored using a vector approach, the usual technique is to apply a boundary model. This means that each area feature is represented by some arc/node structure that determines a polygon as the area's boundary. Common sense dictates that area features of the same kind are best stored Polygons in a single data layer, represented by mutually non-overlapping polygons. In essence, what we then get is an application-determined (i.e. adaptive) partition of space.

Observe that a polygon representation for an area object is yet another example of a finite approximation of a phenomenon that inherently may have a curvi-linear boundary.

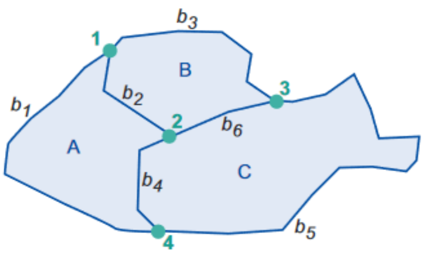


A simple but naive representation of area features would be to list for each polygon simply the list of lines that describes its boundary. Each line in the list would, as before, be a sequence that starts with a node and ends with one, possibly with vertices in between.

The line that makes up the boundary between them is the same, which means that using the above representation the line would be stored twice, namely once for each polygon. This is a form of data duplication—known as data redundancy—which is (at least in theory,) unnecessary, although it remains a feature of some systems. There is another disadvantage to such polygon-by-polygon representations. If we want to find out which polygons border the bottom left polygon, we have to do a rather complicated and time-consuming analysis comparing the vertex lists of all boundary lines with that of the bottom left polygon.

The boundary model is an improved representation that deals with these disadvantages. It stores parts of a polygon's boundary as non-looping arcs and indicates which polygon is on the left and which is on the right of each arc. A simple example of the boundary model is provided in Figure. It illustrates which additional information is stored about spatial relationships between lines and polygons. Obviously, real coordinates for nodes (and vertices) will also be stored in another table.

line	*from*	*to*	*left*	*right*	*vertexlist*
b_1	4	1	W	A	...
b_2	1	2	B	A	...
b_3	1	3	W	B	...
b_4	2	4	C	A	...
b_5	3	4	W	C	...
b_6	3	2	C	B	...



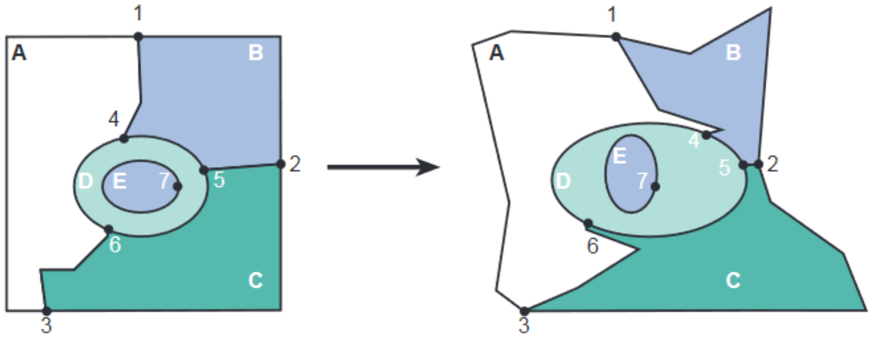
2.3.4 Topology and spatial relationships

General spatial topology:

Topology deals with spatial properties that do not change under certain transformations. For example, features drawn on a sheet of rubber can be made to change in shape and size by stretching and pulling the sheet. However, some properties of these features do not change:

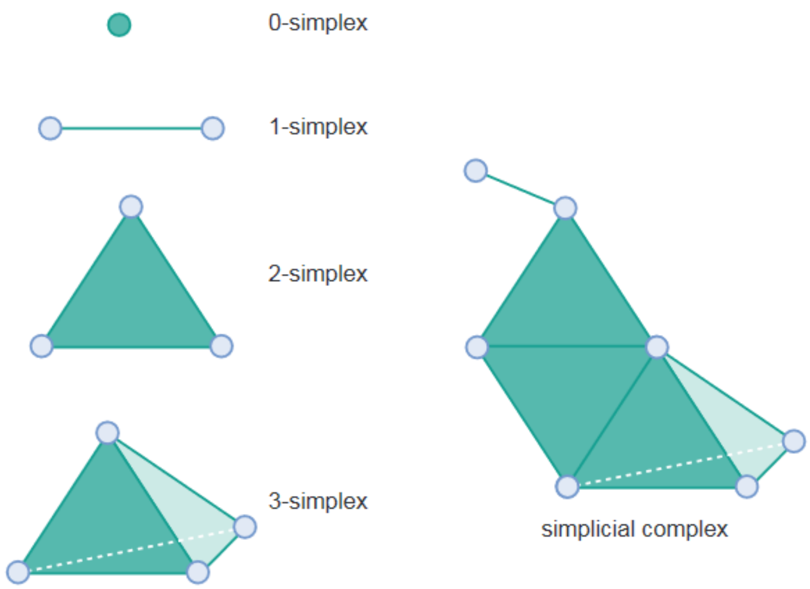
•Area E is still inside area D,

•The neighbourhood relationships between A,B,C,D, and E stay intact, and their boundaries have the same start and end nodes, and
•The areas are still bounded by the same boundaries, only the shapes and lengths of their perimeters have changed.



Topological relationships are built from simple elements into more complex elements: nodes define line segments, and line segments connect to define lines, which in turn define polygons.
In what follows below, we will look at aspects of topology in two ways. Firstly, using simplices, we will look at how simple elements (points) can be combined to define more complex ones (lines and polygons). Secondly, we will examine the logical aspects of topological relationships using set-theory.

Topological relationships:
The mathematical properties of the geometric space used for spatial data can be described as follows:
•The space is a three-dimensional Euclidean space where for every point we can determine its three-dimensional coordinates as a triple (x,y,z) of real numbers. In this space, we can define features like points, lines, polygons, and volumes as geometric primitives of the respective dimension. A point is zero-dimensional, a line one-dimensional, a polygon two-dimensional, and a volume is a three-dimensional primitive.
•The space is a metric space, which means that we can always compute the distance between two points according to a given distance function. Such a function is also known as a metric.
•The space is a topological space, of which the definition is a bit complicated. In essence, for every point in the space we can find a neighbourhood around it that fully belongs to that space as well.
•Interior and boundary are properties of spatial features that remain invariant under topological mappings. This means, that under any topological mapping, the interior and the boundary of a feature remains unbroken and intact.



There are a number of advantages when our computer representations of geographic phenomena have built-in sensitivity of topological issues. Questions related to the 'neighbourhood' of an area are a point in case. To obtain some 'topological sensitivity' simple building blocks have been proposed with which more complicated representations can be constructed:

•We can define within the topological space, features that are easy to handle and that can be used as representations of geographic objects. These features are called simplices as they are the simplest geometric shapes of some dimension: point (0-simplex), line segment (1-simplex), triangle (2-simplex), and tetrahedron (3-simplex).

•When we combine various simplices into a single feature, we obtain a simplicial complex.

The topology of two dimensions:

Suppose we consider a spatial region A. It has a boundary and an interior, both seen as (infinite) sets of points, and which are denoted by boundary(A) and interior(A), respectively. We consider all possible combinations of intersections(∩) between the boundary and the interior of A with those of another region B, and test whether they are the empty set (∅) or not. From these intersection patterns, we can derive eight (mutually exclusive) spatial relationships between two regions. If, for instance, the interiors of A and B do not intersect, but their boundaries do, yet a boundary of one does not intersect the interior of the other, we say that A and B meet.

$$
\begin{aligned}
A\ meets\ B \ \stackrel{\text{def}}{=}\ & interior(A) \cap interior(B) = \emptyset \ \wedge \\
& boundary(A) \cap boundary(B) \neq \emptyset \ \wedge \\
& interior(A) \cap boundary(B) = \emptyset \ \wedge \\
& boundary(A) \cap interior(B) = \emptyset.
\end{aligned}
$$

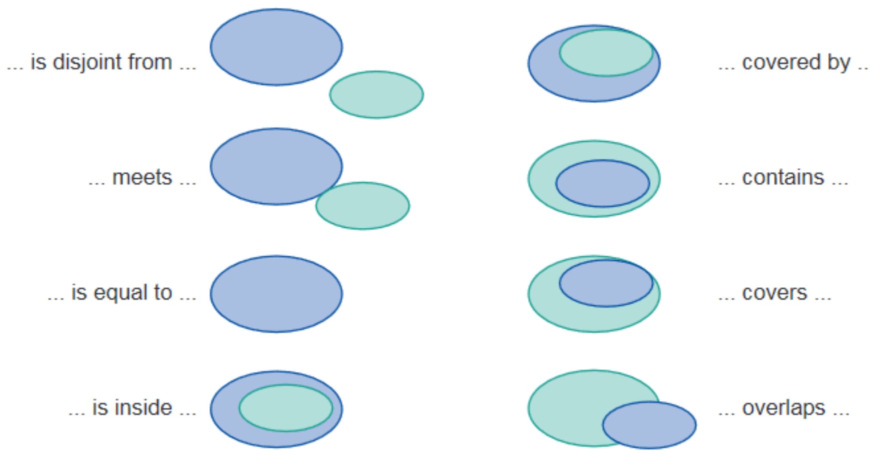
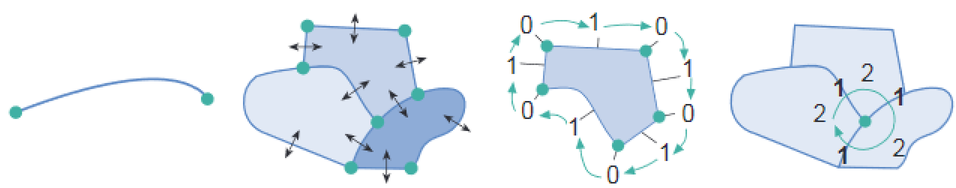


Figure shows all eight spatial relationships: disjoint, meets, equals, inside, covered by, contains, covers, and overlaps. These relationships can be used in queries against a spatial database, and represent the 'building blocks' of more complex spatial queries.

A set of rules defines the topological consistency of that space:

1. Every 1-simplex ('arc') must be bounded by two 0-simplices ('nodes', namely its begin and end node)
2. Every 1-simplex borders two 2-simplices ('polygons', namely its 'left' and 'right' polygons)
3. Every 2-simplex has a closed boundary consisting of an alternating (and cyclic) sequence of 0- and 1-simplices.
4. Around every 0-simplex exists an alternating (and cyclic) sequence of 1- and 2-simplices.
5. 1-simplices only intersect at their (bounding) nodes.



rule (1) rules (2, 5) rule (3) rule (4)

The three-dimensional case:

Many application domains make use of elevational, but these are usually accommodated by so-called 2(1/2)D data structures. These 2(1/2)D data structures are similar to the (above discussed) 2Ddata structures using points, lines and areas.

Solid representation is an important feature for some dedicated GIS application domains. Two of these are worth mentioning here: mineral exploration, where solids are used to represent ore bodies, and urban models, where solids may rep-resent various human constructions like buildings and sewer canals.

A solid can be defined as a true 3D object. An important class of solids in 3D GIS is formed by the polyhedra, which are the solids limited by planar facets. A facet is polygon-shaped, flat side that is part of the boundary of a polyhedron.

2.3.5 Scale and resolution:

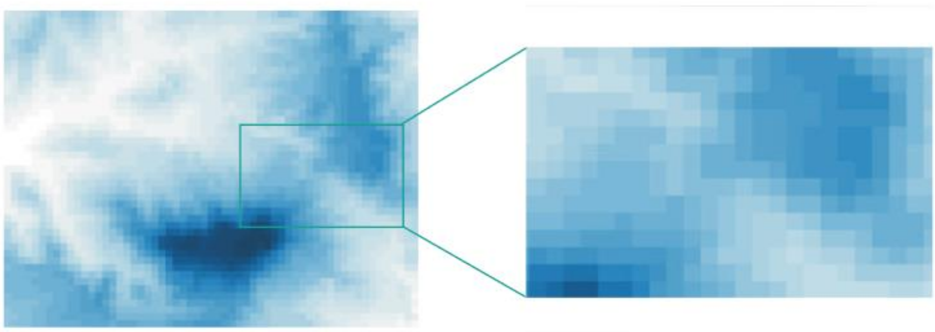
Map scale can be defined as the ratio between the distance on a paper map and the distance of the same stretch in the terrain. A 1:50,000 scale map means that1 cm on the map represents 50,000 cm, i.e. 500 m, in the terrain. ‘Large-scale’ maps means that the ratio is large, so typically it means there is much detail, as in a 1:1,000 paper map. ‘Small-scale’ in contrast means a small ratio, hence less detail, as in a 1:2,500,000 paper map. Digital spatial data, as stored in a GIS, is essentially without scale: scale is a ratio notion associated with visual output, like a map or on-screen display, not with the data that was used to produce the map.

2.3.6 Representations of geographic fields:

A geographic field can be represented through a tessellation, through a TIN or through a vector representation. The choice between them is determined by the requirements of the application at hand. It is more common to use tessellations, notably rasters, for field representation, but vector representations are in use too. We have already looked at TINs.

Raster representation of a field:

Different shades of blue indicate different elevation values, with darker blues indicating higher elevations. The choice of a blue colour spectrum is only to make the illustration aesthetically pleasing; real elevation values are stored in the raster, so instead we could have printed a real number value in each cell.



A raster can be thought of as a long list of field values: actually, there should be m×n such values. The list is preceded with some extra information, like a single georeference as the origin of the whole raster, a cell size indicator, the integer values form and n, and a data type indicator for interpreting cell values. Rasters and quad trees do not store the georeference of each cell, but infer it from the above information about the raster.

Vector representation of a field:

We briefly mention a final representation for fields like elevation, but using a vector representation. This technique uses isolines of the field. An isoline is a linear feature that connects the points with equal field value. When the field is Isoline elevation, we also speak of contour lines.

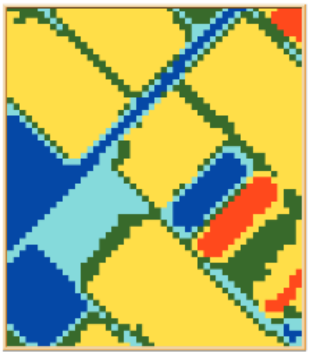
Isolines as are presentation mechanism are not very common, however. They are in use as a geo-information visualization technique (in mapping, for instance), but commonly using a TIN for representing this type of field is the better choice. Many GIS packages provide functions to generate a isoline visualization from a TIN.

2.3.7 Representation of geographic objects:

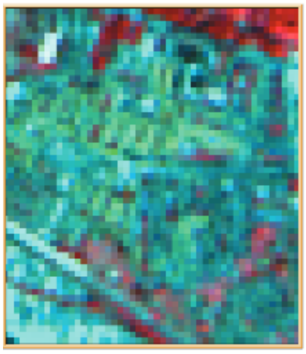
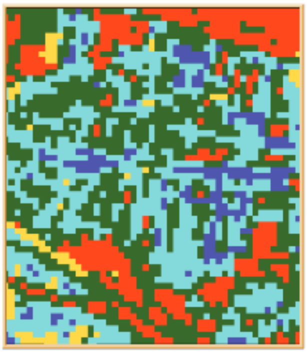
The representation of geographic objects is most naturally supported with vectors. After all, objects are identified by the parameters of location, shape, size and orientation, and many of these parameters can be ex-pressed in terms of vectors.

Tessellations to represent geographic objects:

Remotely sensed images are an important data source for GIS applications. Un-processed digital images contain many pixels, with each pixel carrying a reflectance value. Various techniques exist to process digital images into classified images that can be stored in a GIS as a raster. Image classification attempts Image classification to characterize each pixel into one of a finite list of classes, thereby obtaining an interpretation of the contents of the image. Image classification attempts to characterize each pixel into one of a finite list of classes, thereby obtaining an interpretation of the contents of the image.

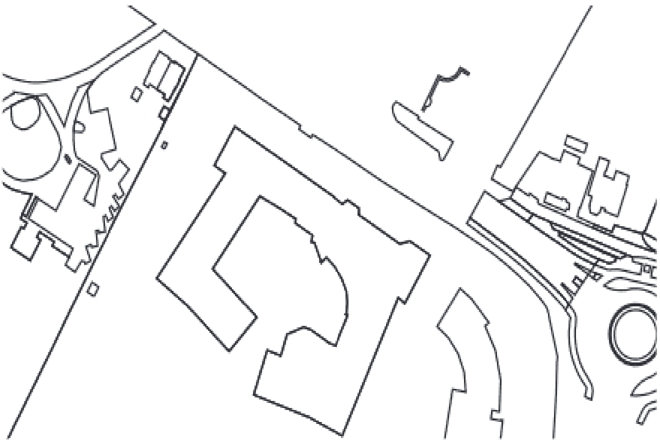
- Area objects are conveniently represented in raster, area boundaries may appear as jagged edges.
- Line and point objects are not convenient to represent using raster.

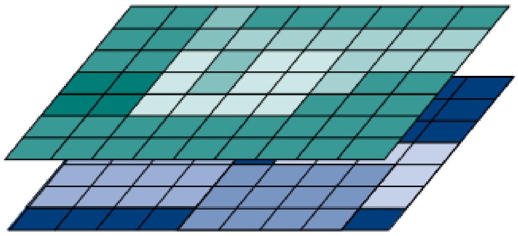
Line and point objects are more awkward to represent using rasters. After all, we could say that rasters are area-based, and geographic objects that are perceived as lines or points are perceived to have zero area size. Standard classification techniques, moreover, may fail to recognize these objects as points or lines.

Vector representations for geographic objects:

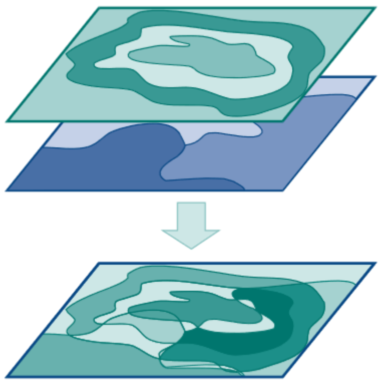
The somehow more natural way to represent geographic objects is by vector representations.



2.4 Organizing and managing spatial data:



- A spatial data layer is either a representation of a continuous or discrete field, or a collection of objects of the same kind.
- The data is organized so that similar elements are in a single data layer. For example, all telephone booth point objects would be in one layer, and all road line objects in another.
- Data layers can be overlaid with each other, inside the GIS package, so as to study combinations of geographic phenomena.
- The spatial relationships between different phenomena, require computations which overlay one data layer with another.



2.5 The temporal dimension:

Geographic phenomena are also dynamic; they change over time.

Examples of the kinds of questions involving time include:

- Where and when did something happen?
- How fast did this change occur?
- In which order did the changes happen?

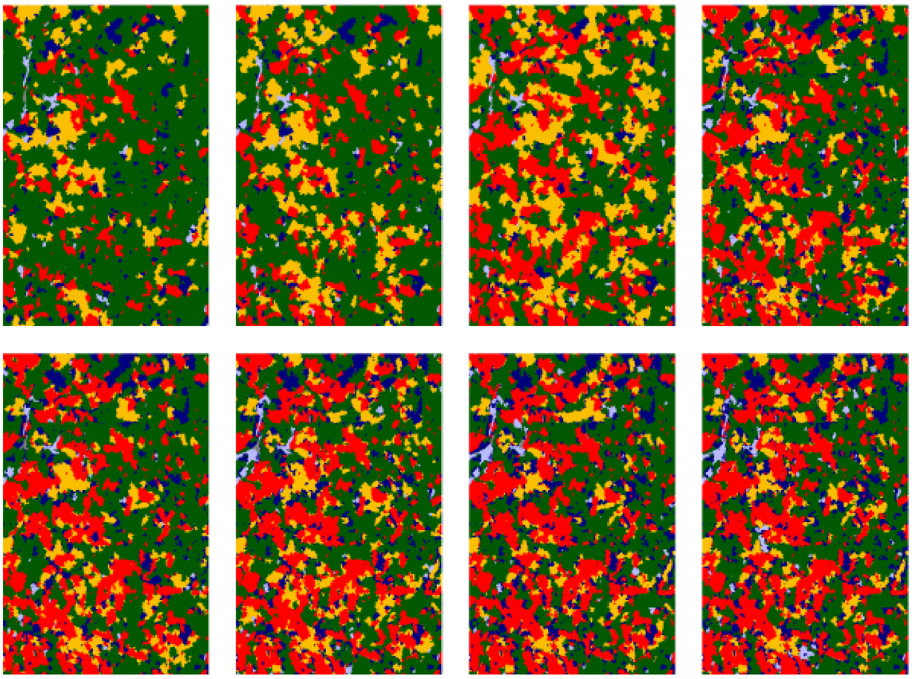
Representing time in GIS:

- Spatiotemporal data models are ways of organizing representations of space and time in a GIS.
- The most common of these is a ‘snapshot’ state that represents a single point in time of an ongoing natural or man-made process.
- We may store a series of these snapshot states to represent change

Different ‘concepts’ of time:

- Discrete and continuous time : Time can be measured along a discrete or continuous scale.

- Discrete time is composed of discrete elements (seconds, minutes, hours, days, months, or years).
- In continuous time, no such discrete elements exist, and for any two different points in time, there is always another point in between. Derive temporal relationships between events and periods such as ‘before’, ‘overlap’, and ‘after’.
- Valid time and transaction time: Valid time (or world time) is the time when an event really happened, or a string of events took place. Transaction time (or database time) is the time when the event was stored in the database or GIS.
- Linear, branching and cyclic time: Time can be considered to be linear, extending from the past to the present (‘now’), and into the future. Branching time—in which different time lines from a certain point in time onwards are possible—and cyclic time—in which repeating cycles such as seasons or days of a week are recognized.
- Time granularity: When measuring time, granularity is the precision of a time value in a GIS or database (e.g. year, month, day, second, etc.). Different applications may obviously require different granularity.
- Absolute and relative time: Time can be represented as absolute or relative. Absolute time marks a point on the time line where events happen (e.g. ‘6 July 1999 at 11:15 p.m.’). Relative time is indicated relative to other points in time (e.g. ‘yesterday’, ‘last year’, ‘tomorrow’, which are all relative to ‘now’, or ‘two weeks later’.
- Change detection: Studies of this type are usually based on some ‘model of change’, which includes knowledge and hypotheses of how change occurs for the specific phenomena being studied. It includes knowledge about speed of tree growth.
- Spatiotemporal analysis: we consider changes of spatial and thematic attributes over time. We can keep the spatial domain fixed and look only at the attribute changes over time for a given location in space.
- On the other hand, we can keep the attribute domain fixed and consider the spatial changes over time for a given thematic attribute.
- This may lead to notions of **object motion**, a subject receiving increasing attention in the literature. Applications of moving object research include traffic control, mobile telephony, wildlife tracking, vector-borne disease control, and weather forecasting.
- Object identity: When does a change or movement cause an object to disappear and become a new one? With wildlife this is quite obvious.



Chapter 01

A gentle introduction to GIS

1. What are the different applications of GIS?
2. Define GIS. How data capture and preparation is done in GIS?
3. Explain Data manipulation and analysis techniques used in GIS.
4. Differentiate between GISystems and GIScience.
5. What are the key components Data quality parameters?
6. What are maps? How it is used in GIS?
7. What is a database? What are the advantages of using it in GIS?

Chapter 02

Geographic information and Spatial data types

1. Explain how models are used as representation of the real world.
2. Define Geographic phenomena in 2D and 3D.
3. Differentiate between the different types of Geographic fields.
4. What are the different data types used in GIS.
5. Explain Regular Tessellation with its types.
6. What is Irregular Tessellation with reference to quadtree?
7. How TIN is used in representation of objects?
8. Explain Point, Line and Area representation.
9. What are the set of rules to define the topological consistency of that space?
10. Explain the meaning of Scale and resolution.
11. How will you represent Time in GIS?

Multiple choice questions:

1. A ___________ might be interested in the impact of slash-and-burn practices on the populations of amphibian species in the forests of a mountain range to obtain a better understanding of long-term threats to those populations.

A. urban planner
B. natural hazard
C. biologist
D. geological engineer

ANSWER: C

2. A ___________ could be interested in determining which prospective copper mines should be selected for future exploration, taking into account parameters such as extent, depth and quality of the ore body, amongst others.

A. urban planner
B. mining engineer
C. biologist
D. geological engineer

ANSWER: B

3. __________________ is the scientific field that attempts to integrate different disciplines studying the methods and techniques of handling spatial information.

A. Geo-Information Science
B. GISystems
C. GIS applications
D. None of these

ANSWER: A

4. _________________ is responsible for collecting topographic data for the entire country following pre-set standards.

A. GIS
B. GIS System
C. None of these
D. National Mapping Agencies (NMAs)

ANSWER: D

5. A _______________ is a miniature representation of some part of the real world.

A. Model
B. Map

C. All of these
D. Digitized map
ANSWER: C

6. _____________ is the science and art of map making, functions as an interpreter, translating real world phenomena (primary data) into correct, clear and understandable representations for our use.
A. Cryptography
B. Cartography
C. Map making
D. None of these
ANSWER: B

7. A ___________ is a repository for storing large amounts of data.
A. column
B. Record
C. Table
D. Database
ANSWER: D

8. What are the functions of a Database?
A. A database can be used by single user at a time.
B. A database offers a number of techniques for storing data and allows the use of the most efficient one—i.e. it supports storage optimization.
C. A database allows the imposition of rules on the stored data; rules that will not be automatically checked after each update to the data—i.e. it does not support data integrity.
D. None of these
ANSWER: B

9. ______________ can be informally defined as a model of space in which locations are represented by coordinates—(x,y) in 2D; (x,y,z) in 3D—and distance and direction can defined with geometric formulas.
A. Euclidean space
B. Disk Space
C. None of these
D. 3D Space
ANSWER: A

10. In a ______________ field, the underlying function is assumed to be 'mathematically smooth, meaning that the field values along any path through the study area do not change abruptly, but only gradually.
A. middle

B. discrete
C. continuous
D. None of these
ANSWER: C

11. ____________ are values that provide a name or identifier so that we can discriminate between different values.
A. Ratio data values
B. Interval data values
C. Ordinal data values
D. Nominal data
ANSWER: D

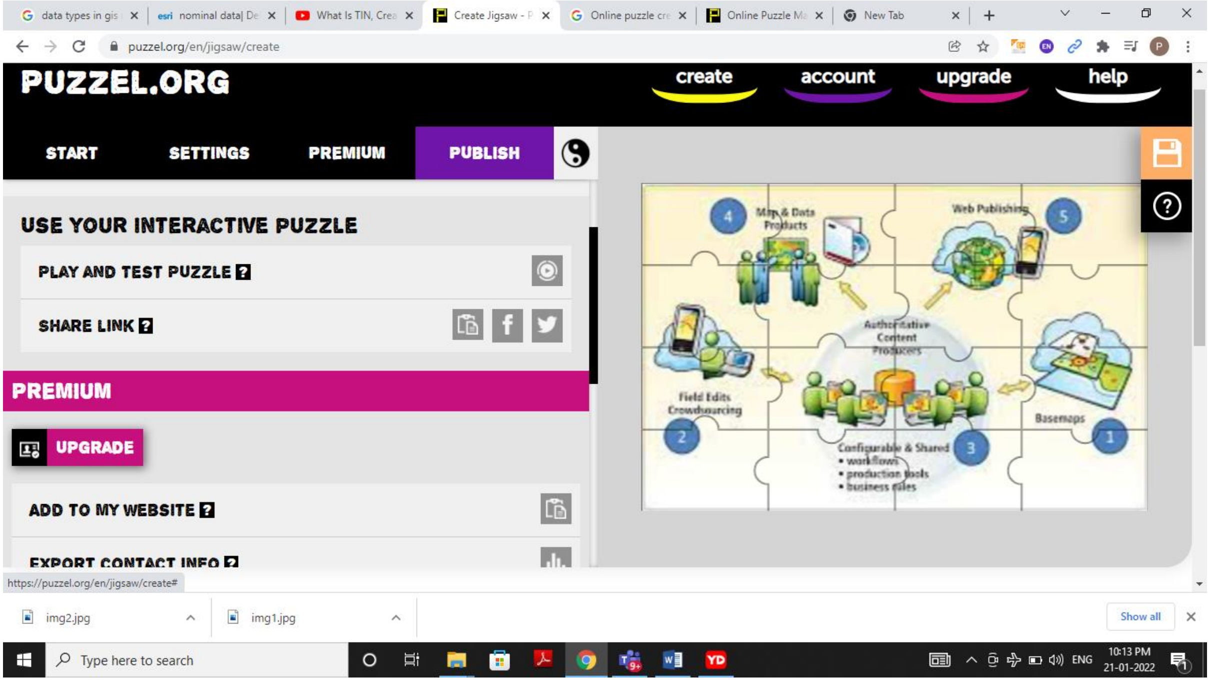
12. ____________ are data values that can be put in some natural sequence but that do not allow any other type of computation.
A. Ratio data values
B. Interval data values
C. Ordinal data values
D. Nominal data
ANSWER: C

13. Geographic objects are usually easily distinguished and named, and their position in space is determined by a combination of one or more of the following parameters:
A. Location
B. Shape
C. Size
D. All of these
ANSWER: D

14. In a ____________ tessellation, the cells are the same shape and size.
A. regular
B. irregular
C. TIN
D. None of these
ANSWER: A

15. Quad tree is an example of ____________ tessellation.
A. regular
B. irregular
C. TIN
D. None of these
ANSWER: B

Additional Activities:



Click on the link and solve the puzzle:

https://puzzel.org/en/jigsaw/play?p=-MtxWQooeniDn_ns8mQt

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